

Kyle Main

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Data Engineer — Large-Scale Pipelines, Analytics Infrastructure & Abuse-Detection Data Systems

Data engineer with 12 years of experience designing and running the pipelines, warehousing, and analytics infrastructure that power large-scale detection and safety systems — including ETL/ELT ownership for 220+ ingested log/data sources, a company-wide Common Information Model standardizing schema across all of it, and Apache Beam programs run on GCP Dataflow for large-scale historical data retrieval. That foundation sits directly under years of hands-on abuse/threat detection content development and threat-intelligence integration — signature, statistical, behavioral, and ML-based detection logic built on the pipelines and data models I engineered, plus dashboarding and data-quality alerting for stakeholder self-service.

CORE SKILLS

Data Pipelines & ETL/ELT: Pipeline design and ownership for 220+ ingested log/data sources; Common Information Model (CIM) / data dictionary design standardizing field names and types across all ingested data; 50+ Logstash parsing/normalization filters; Apache Beam programs run via GCP Dataflow for large-scale historical/cold-storage data retrieval; connector/collector health monitoring and troubleshooting

Cloud Data Platforms: GCP (BigQuery, Dataproc, Dataflow, serverless event-driven enrichment), AWS, Azure; Elasticsearch as a large-scale analytical data store (Query DSL, transforms, Beats-based ingestion, API)

Distributed Data Processing & Streaming: PySpark / SparkSQL for large-scale exploratory data analysis on GCP Dataproc clusters with Zeppelin notebooks; familiarity with Kafka and Flink for event-streaming workloads

Dashboards, Reporting & Data Quality: Custom Kibana dashboards/visualizations on Elasticsearch transforms for stakeholder self-service reporting; data-quality monitoring and alerting to keep safety-critical data reliable

Abuse & Threat Detection Data Systems: Detection content development (signature, statistical, behavioral, ML-based) on engineered data pipelines — 2,300+ detection rules across the MITRE ATT&CK matrix; threat-intelligence integration into detection logic and alert enrichment

Languages & Data Science: Python, SQL, Pandas, scikit-learn, NumPy; clustering/unsupervised ML; time-series anomaly detection; Git

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Built an entirely new security data ingestion platform for DOE/NSA from the ground up — CrowdStrike, Suricata, and Zeek telemetry into a central Elasticsearch environment — with data transforms, a UEBA detection layer, data-quality alerting, and custom Kibana dashboards (DOE/NSA Security Data Integration project, completed).
- Supported data ingestion and data-quality efforts within an Elasticsearch/Splunk environment for DOE's Continuous Diagnostics and Mitigation (CDM) program, a federal continuous-monitoring initiative (CISA CDM at DOE, completed).
- Currently create and manage detection/alerting analytics (Splunk saved searches) directly on top of ingested SOC data supporting Treasury's Security Operations Center incident response and case work (Treasury SOC / TSSOC, current project).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and detection engineering team building signature, behavioral, statistical, and ML-based detection content against massive-scale customer telemetry on a cloud-based big-data platform, incorporating threat-intelligence context from Forescout's in-house research team (Vedere Labs) into detection logic and alert enrichment.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Owned data engineering for a next-gen cloud SIEM at massive scale as a very early hire — 220+ ingested log sources, 50+ Logstash parsing/normalization filters, and a Common Information Model standardizing field names/types across all of it — plus connector/collector health monitoring and a homegrown Apache Beam program run via GCP Dataflow to fetch large volumes of historical cold-storage data for customers.
- Ran exploratory data analysis at scale on GCP Dataproc clusters using Zeppelin notebooks and a homegrown analysis toolkit — Spark jobs loading data from buckets, PySpark/SparkSQL analysis feeding detection-rule development.
- Created and managed 2,300+ individual detection rules covering most of the MITRE ATT&CK matrix, building out the rules engine and detection content for the startup from scratch, with threat intelligence integrated into rule logic and used directly for false-positive investigation research.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Analyzed large-scale security log data to build custom detection models — DNS-based malware detection/mitigation and anomalous-behavior discovery across the network.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox